

Arjuna JEE (2025)

Maths

DPP: 8

Sequence & Series

- Q1** If a, b, c are positive numbers and $abc = 1$ then minimum value of $a + b + c$ is
- Q2** Suppose $\log_a b + \log_b a = c$. Then the smallest possible integer value of c for all, $a, b > 1$ is
(A) 4 (B) 3
(C) 2 (D) 1
- Q3** If $a_i > 0, i = 1, 2, 3, \dots, 50$ and $a_1 + a_2 + \dots + a_{50} = 50$ then the minimum value of $\frac{1}{a_1} + \frac{1}{a_2} + \frac{1}{a_3} + \dots + \frac{1}{a_{50}}$ is
(A) 150 (B) 100
(C) 50 (D) $(50)^2$
- Q4** The minimum value of $\frac{x}{y} + \frac{y}{x} + \frac{y}{z} + \frac{z}{y} + \frac{x}{z} + \frac{z}{x}$ where x, y, z all are +ve is
(A) 4 (B) 6
(C) 12 (D) 8
- Q5** If three positive numbers a, b, c are in AP such that $abc = 8$ then the minimum possible value of b is
(A) 2
(B) $4^{\frac{1}{3}}$
(C) $4^{\frac{2}{3}}$
(D) 4
- Q6** If $a_1, a_2, a_3, \dots, a_{20}$ are AMs between 13 and 67 then the maximum value of $a_1, a_2, a_3, \dots, a_{20}$ is
(A) $(20)^{20}$
(B) $(40)^{20}$
(C) $(60)^{20}$
(D) $(80)^{20}$
- Q7** The minimum value of $27 \tan^2 \theta + 3 \cot^2 \theta$ is
(A) 9 (B) 3
(C) 6 (D) 18
- Q8** $0 < x < \frac{\pi}{4}$ then the minimum value of $(\sin x + \cos x + \operatorname{cosec} 2x)^3$ is
(A) 27 (B) 13.5
(C) 6.75 (D) None
- Q9** The minimum value of $x + y$ if x and y satisfy $\log_2 x + \log_2 y \geq 6$ is
(A) 6 (B) 2
(C) 16 (D) 14
- Q10** If $\alpha \in (0, \frac{\pi}{2})$ then the minimum value of $\sqrt{x^2 + x} + \frac{\tan^2 \alpha}{\sqrt{x^2 + x}}$ is
(A) 2
(B) $2 \tan \alpha$
(C) $\tan 2\alpha$
(D) $\tan^2 \alpha$
- Q11** If x, y, z are three positive real number then the minimum value of $(x + y + z) \left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z} \right)$ is
(A) 6 (B) 9
(C) 7 (D) 8
- Q12** The minimum value of the sum of the numbers a^{-5}, a^{-4}, a^9 is
(A) 1 (B) 2
(C) 3 (D) 5
- Q13** The minimum value of $4^x + 4^{1-x}, x \in R$ is
(A) 1 (B) 2
(C) 3 (D) 4
- Q14** The minimum value of the sum of real numbers $a^{-5}, a^{-4}, 3a^{-3}, 1, a^8$ and a^{10} with $a > 0$ is
(A) 6 (B) 7
(C) 8 (D) 9
- Q15** If $S_n = \frac{1}{6.11} + \frac{1}{11.16} + \frac{1}{16.21} + \dots$ Upto n terms, then $6S_n$ equal



- (A) $\frac{5n-4}{5n+6}$
 (B) $\frac{n}{5n+6}$
 (C) $\frac{2n-1}{5n+6}$
 (D) $\frac{1}{5n+6}$

Q16 If

$\sum_{k=1}^n k(k+1)(k-1) = pn^4 + qn^3 + tn^2 + sn$, where p, q, t and s are constants, then the value of s is equal to

- (A) $-\frac{1}{4}$
 (B) $-\frac{1}{2}$
 (C) $\frac{1}{2}$
 (D) $\frac{1}{4}$

Q17 The sum of series $3 + 7 + 13 + 21 + 31 + \dots$ to n terms is equal to

- (A) $\frac{n(n^2+2n+3)}{3}$
 (B) $\frac{n(n^2+3n+5)}{3}$
 (C) $\frac{n(n^2+2n+5)}{3}$
 (D) $\frac{n(n^2+2n+2)}{3}$

Q18 If $1^3 + 2^2 + 3^2 + \dots + 2003^2 = (2003)(4007)(334)$ and
 $(1)(2003) + (2)(2002) + (3)(2001) + \dots + (2003)(1) = (2003)((334)(x))$
 then the value of x is equal to

- (A) 2005 (B) 2004
 (C) 2003 (D) 2001



Answer Key

Q1 3
Q2 (C)
Q3 (C)
Q4 (B)
Q5 (A)
Q6 (B)
Q7 (D)
Q8 (B)
Q9 (C)

Q10 (B)
Q11 (B)
Q12 (C)
Q13 (D)
Q14 (C)
Q15 (B)
Q16 (B)
Q17 (B)
Q18 (A)



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